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NANOMATERIALS

$$1 \text{ nm} = 10^{-9} \text{ m}$$

Material with ^{one or more} ~~the~~ dimension in the range of 0.1 nm to 100 nm, known as nanomaterials. It can be

metal, ceramics, composite, polymeric material, etc.

Nanomaterials are biological, inorganic or organic structure. ^{Due} to their small size, display unique

properties, different from those observed in bulk material.

• opaque to transparent (Cu)

• Inert to catalyst (Pt)

• Stable to combustible (Al)

• Solids to liquids (Au)

• Insulator to conductor (Si)

PROPERTIES

There are two types of properties:

i) Physical property

ii) Chemical property

Physical Properties

i) When material is converted into nanoscale, they show different colour from the normal material.

Eg: gold material is converted to red colour when converted into nanoscale.

ii) When material is converted into nanoscale, its melting point decreases.

Eg: nanowires, nanotubes and nanoparticles.

iii) When material is converted into nanoscale, its crystal perfection increases and defects reduces, so the mechanical strength increases.

iv) When material is converted into nanoscale, its ~~ele~~ electrical resistivity increases.

Chemical Properties.

i) When material is converted into nanoscale, its chemical properties also change like surface area, and its reactivity increases enormously.

CLASSIFICATION

① On the basis of dimensions:

Nanomaterials

0-D

All dimensions are in nanorange.

Eg: nanosilver, nanogold.



1-D

One dimension is out of nanorange.

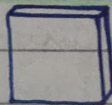
Eg: nanorod, nanotube, nanowires,



2-D

Two dimensions are out of nanorange.

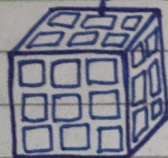
Eg: nanoplates, nanofilms, nanocoatings.



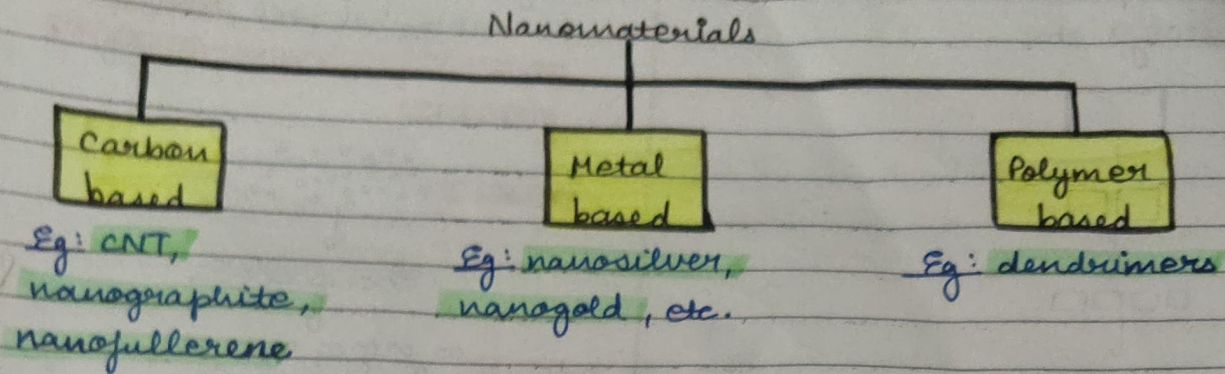
3-D

All dimensions are out of nanorange.

Eg: ~~be~~ multibundles of nanorods, nanowires, nanoplates.



② On the basis of material



METHODS OF SYNTHESIS

There are two approaches:

- 1) Top down approaches
- 2) Bottom up approaches

Top-down approach

* In this approach, the nanoscale material is formed from large substrate (bulk material) by repeated pattern transfer step.

* It is ~~very~~ ^{less} expensive.

* ~~Slow process and not~~ suitable for large scale production. (Fast process)

Bottom-up approach

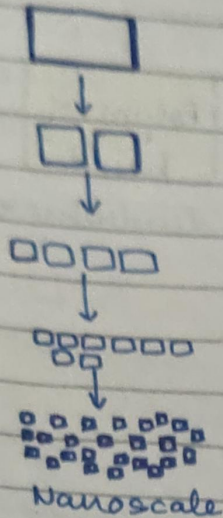
* In this approach, the atoms or molecules combine together to form nanoscale material.

* It is ~~less~~ ^{very} expensive.

* Suitable for ~~large~~ ^{small} scale production. ($\because$ slow process)

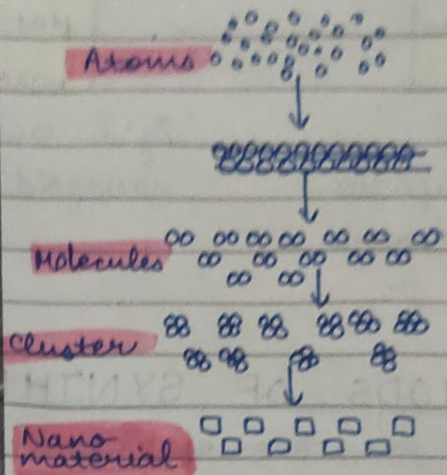
* Heterogeneous (difficult to control shape and size)

* Eg: lithography, etching, mechanical grinding, etc.



* Homogeneous (control over shape and size)

Eg: molecular self assembly, synthesis of 3D protein in biological cell, CVD, ion implantation, sol-gel method.



APPLICATIONS (12)

- i) Nanoscale metals act as a good catalyst.
Eg: MnO_2
 - ii) Mechanical strength of nanomaterial is increased as compared to normal material.
Eg: CNT, bucky ball are extremely strong and used in manufacturing of bullet-proof jacket.
 - iii) The fabrics of nanomaterial enhances and improves the performance as compared to conventional fabrics.
- Organic LEDs (OLEDs) used in displays of new TV, camera, etc.
 - Nanotubes used for storing hydrogen fuel.

RAM (magnetic random access memory)
to make efficient solar cells to produce solar energy.

iv) Nanomaterials are also used in drug delivery system as it decreases the harmful effect of chemotherapy in cancer cell treatment. Eg: dendrimers, CNTs.

v) It is also used in anti-ageing cream and, sunscreens and other cosmetics. Eg: nanofullerenes used in anti-ageing creams.

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vi) Nano-biosensors are used to detect pathogens in food.

vii) Nanomaterials are also used in water purification system. Eg: Nano AgNO_3 and nano-gold are useful to remove the harmful elements ^{like} ~~are~~ arsenic from water.

viii) Nanoparticles are also used in anti-bacterial silver coating on wound healing.

CARBON NANOTUBE (CNT)

Carbon nanotubes are sheets of graphite rolled up to make a tube-like structure known as carbon nanotube. Its diameter is 0.4 nm. It can be single walled

CNT or multi-walled CNT.

(multiple concentric nanotubes
outer layer to add functionality)

(rolling single layer
of graphite, called
graphene layer)

Properties

1) They have exceptionally high tensile strength.

2) They can be used as ~~insulator~~ ^{conductor}, semiconductor and insulator depending on the way the graphite sheets spiral around the tube.

- 3) They possess large surface area.
- 4) They are reported to be thermally stable in a vacuum upto 2800°C .
- 5) They exhibit excellent electrical and mechanical properties.
- 6) They have capacity to carry an electric current 1000 times greater than copper wire.

Applications

- 1) CNT is used as needle to bring active agents from living cells. This process could be applied as a new way to therapy against cancer cell.
- 2) Due to excellent electrical, heat transmission capacity, heat stability, low density and high strength, it is used as fillers in composite material.
- 3) It is also used as electrical energy storage application.